



ROOT CAUSE ANALYSIS

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

RCA — VCF Operations Integration Adapter Warnings

Root cause analysis of integration adapter warnings: NSX credential failures and vCenter STS authentication errors in a VCF 9.0.1 environment.

VCF OPS

INTEGRATION

RCA

NSX/VCENTER

VCF 9.0.1

VMware Cloud Foundation

RCA — VCF Operations Integration Adapter Warnings

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Root Cause Analysis Report

VCF Operations Integration Adapter Warning Status,
NSX Credential Failures, and vCenter STS Authentication Error

Field	Value
Document ID	RCA-2026-0331-001
Date of Incident	March 31, 2026
Date of Report	April 01, 2026
Severity	Warning / Service Degradation
Environment	VCF 9.0.1 Lab -- lab.local
Affected Systems	VCF Operations 9.0.2, NSX Manager (nsx-vip.lab.local), vCenter Server (vcenter.lab.local)
Author	Virtual Control LLC
Status	Resolution in Progress

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1. Executive Summary

On March 31, 2026, all VMware Cloud Foundation adapter instances in VCF Operations (Administration > Integrations > Accounts) were observed in Warning status. The warning affected five adapter instances under the VMware Cloud Foundation account: lab (VCF Adapter), mgmt (two instances), mgmt - vSAN, and nsx-vip.lab.local. The OS and Application Monitoring adapter was unaffected and showed Collecting status.

Investigation revealed three distinct but related root causes:

- ❖ NSX adapter credential failure -- The NSX Manager admin credentials stored in VCF Operations were incorrect, returning HTTP 403 on API calls. The NSX credential entry was missing entirely from VCF Operations and had to be added.
- ❖ NSX service account password expiration -- The NSX audit account password had expired. The root account was also disconnected. Both required password resets on the NSX Manager CLI before VCF Operations could reconnect.
- ❖ vCenter STS authentication error -- The vCenter Server vSphere Client displayed an 'Invalid token' error from the Security Token Service (STS). The STS_INTERNAL_SSL_CERT VECS store was missing. The STS certificates in LDAP were valid (expiry 2036). The STS service required a restart.

The NSX adapter warnings were resolved by resetting passwords, activating expired accounts, and adding the correct credentials to VCF Operations. The vCenter STS issue was addressed by restarting the STS service after confirming the underlying certificates were valid.

2. Environment Details

Component	FQDN / Identifier	Version	Role
SDDC Manager and vCenter Server SSO	sddc-manager.lab.local	VCF 9.0.1	VCF lifecycle management orchestration
NSX Manager VIP	nsx-vip.lab.local	vCenter (VCF 9.0.1 managed)	VM and host management, authentication
NSX Manager Node nsx-node1 (behind VIP)		NSX (VCF 9.0.1 managed)	Network virtualization and micro-segmentation
VCF Operations and analytics	Collector: collector.lab.local	NSX (VCF 9.0.1 managed)	NSX Manager cluster member
VCF Operations for and alerting Logs	Remote Collector: VCF Operations Collector-vcf-ops	9.0.2	Infrastructure monitoring
Domain	(integrated with VCF Operations)	9.0.x	Centralized log management
	SSO: vsphere.local DNS: lab.local	--	Identity and DNS domain

Adapter Inventory at Time of Incident

The following table shows the exact state of the Integrations page (Administration > Integrations > Accounts tab) as observed at the start of troubleshooting:

Account Type	Name	Status	Description	Collector
Version	lab	Warning	VMware Cloud Foundation Adapter	collector.lab.local
--				
VMware Cloud Foundation vcf-op -- (child)	mgmt	Warning	Foundation Adapter	VCF Operations Collector-
9.0.2 (child)	mgmt	Warning	Current instance	s
--				
vcf-op -- (child)	mgmt - vSAN	Warning		VCF Operations Collector-
--				s

(child) vcf-op	nsx-vip.lab.local	Warning	VCF Operations Collector-
OS and Applicati on Monitoring	Application Monitoring Adapter (collector.lab.loc al)	Collecting	s collector.lab.local

3. Issue Description

Primary Symptom

All five VMware Cloud Foundation adapter instances on the VCF Operations Integrations page (Administration > Integrations > Accounts) displayed Warning status with yellow triangle icons. The OS and Application Monitoring adapter was unaffected.

Impact

- ◆ VCF Operations could not collect current data from the NSX Manager, resulting in stale or missing network virtualization metrics.
- ◆ Dashboards and alerts relying on NSX data (such as distributed firewall rules, logical switch health, and load balancer metrics) were not updating.
- ◆ The parent VMware Cloud Foundation adapter reported Warning because one or more child adapters (specifically the NSX adapter) were in a failed state. This cascaded the Warning status to all sibling adapters in the VCF account.
- ◆ Separately, the vCenter Server vSphere Client was returning an authentication error ('Invalid token') which prevented certain backend API operations.

Scope

The issue was confined to the lab.local management domain. The underlying ESXi hosts and virtual machines continued to run normally. vCenter core management functions (VM power operations, vMotion, HA) were not affected. The issue was limited to monitoring data collection (VCF Operations) and vSphere Client backend authentication (STS).

4. Timeline of Events

All times are UTC on March 31, 2026.

Time (UTC)	Action	Result
~12:00	Observed VCF Operations Integrations page. All VMware Cloud adapter as disconnected. OS	Identified nsx-vip.lab.local
~12:02	Foundation adapters showing Warning status. adapter healthy (Collecting).	and Application Monitoring
~12:03		
~12:03	Verified connectivity from collector to SDDC Manager: curl -k (nginx). SDDC Manager https://sddc-manager.lab.local	HTTP 301 Moved Permanently reachable.
~12:05		
~12:06	Verified connectivity from collector to NSX Manager: curl -k reachable (web https://nsx-vip.lab.local	Empty response -- NSX Manager server responding).
~12:08		
~12:09	Tested NSX API authentication: curl -k were incorrect or the https://nsx-vip.lab.local/api/v1/cluster/status locked.' Confirmed	HTTP 403: 'The credentials account specified has been credential failure.
~12:10		
~12:12		
~12:13	Connected to NSX Manager CLI via SSH. Checked admin Account not locked.	admin status = ACTIVE.
~12:15	account status: get user admin status	
~12:18		
~12:20	Updated NSX admin password in VCF Operations Integrations Adapter partially	Admin credential updated.

~12:21	UI (Administration > Integrations > Accounts > Edit accounts still disconnected.	reconnected. audit and root
~12:22	nsx-vip.lab.local)	
~12:23		
~12:24	Identified NSX audit account password as expired. PASSWORD_EXPIRED	audit account status =
~12:25	Reset audit password on NSX CLI: set user audit password successfully.	New password set
	Extended audit password expiration: set user audit password-expiration 9999	Expiration extended.
	Verified audit status: get user audit status PASSWORD_EXPIRED despite	Status still showed
	yet activated.	password reset. Password not
	Exited NSX CLI. From collector, initiated SSH login as audit activation. Audit account user: ssh audit@nsx-vip.lab.local	SSH login forced password became active.
	Reset root password on NSX CLI similarly. (admin, audit, root) now active.	All three NSX accounts
	Added NSX credentials (admin, audit, root) to VCF Operations. Administration > Integrations > Credentials were missing entirely -- not just incorrect. button.	Credentials added via Accounts > CREDENTIALS
	Encountered separate vCenter error: 'Could not authenticate to identified. backend service: Invalid token' in vSphere Client.	STS authentication failure
Valid.	SSH to vCenter. Checked trusted certificates: dir-cli trustedcert list	1 trusted certificate found.
	Checked STS_INTERNAL_SSL_CERT VECS store: vecs-cli ERROR_OBJECT_NOT_FOUND. Store entry list --store STS_INTERNAL_SSL_CERT	Error 4312: missing.
	Checked MACHINE_SSL_CERT VECS store. 2026, Not After Feb 7, 2028.	Valid: Not Before Feb 7,
	Listed all VECS stores: vecs-cli store list present in store list.	STS_INTERNAL_SSL_CERT not Confirmed missing.
	Verified all vCenter services running: service-control --status --all All critical services running including vmware-std.	
	Checked STS certificates via LDAP: sts-cert-support.sh --help (ssoserverSign) and CA cert both	STS signing cert present in LDAP. Valid until
	2036.	
~12:27	Attempted STS restart: vmon-cli --restart sts Restart initiated.	

5. Root Cause Analysis

Three distinct root causes were identified. They are presented in the order they were discovered during troubleshooting.

5.1 Issue 1: NSX Adapter Credential Failure (HTTP 403)

Symptom:

The nsx-vip.lab.local adapter in VCF Operations showed Warning status. The parent VMware Cloud Foundation adapter (lab) and all sibling adapters (mgmt, mgmt - vSAN) also showed Warning as a cascade effect.

Root Cause:

The NSX Manager API returned HTTP 403 ('The credentials were incorrect or the account specified has been locked') when VCF Operations attempted to authenticate. The NSX admin credential was not stored in VCF Operations at all -- it was missing from the credential store, not merely incorrect.

Evidence:

From the VCF Operations collector appliance:

```
curl -k https://nsx-vip.lab.local/api/v1/cluster/status
```

Response:

```
{"module_name":"common-services","error_message":"The credentials were incorrect or the account specified has been locked.","error_code":403}
```

Cascade Effect:

In VCF Operations, the VMware Cloud Foundation adapter is a parent account that manages child adapters for vCenter (mgmt), vSAN (mgmt - vSAN), and NSX (nsx-vip.lab.local). When any child adapter enters a Warning or failed state, the parent adapter also reports Warning. This caused all five entries to display Warning even though only the NSX adapter had a direct failure.

5.2 Issue 2: NSX Service Account Password Expiration

Symptom:

After updating the NSX admin credential in VCF Operations, the NSX adapter partially reconnected. However, the audit and root service accounts remained disconnected.

Root Cause:

The NSX audit account password had expired. NSX Manager has a password expiration policy for local accounts. When the password expires, the account enters PASSWORD_EXPIRED status and all API and SSH authentication attempts fail, even with the correct password.

Evidence:

From the NSX Manager CLI (nsx-node1):

```
nsx-node1> get user audit status
```

Output:

```
Username: audit Status: PASSWORD_EXPIRED
```

Complication:

After resetting the audit password via set user audit password on the NSX CLI, the status still showed PASSWORD_EXPIRED. Setting the password from the admin CLI session was not sufficient to activate the account. The audit user had to perform an interactive SSH login (ssh audit@nsx-vip.lab.local) to force password activation. This is a known NSX behavior where expired accounts require an interactive login to complete the password change workflow.

Additional Finding:

The root account on NSX Manager does not support the get user root status command. Root is managed differently on the NSX appliance. The root account was also disconnected and required a password reset.

```
nsx-node1> get user root status
% Operation is not valid for root user
```

5.3 Issue 3: vCenter STS Authentication Error (Invalid Token)

Symptom:

The vSphere Client displayed the following error:

```
Error: Could not authenticate to backend service: java.util.concurrent.CompletionException:
com.vmware.vcenter.apigw.security.AuthenticationException: Failed to log into
[uri=http://localhost:1080/external-vecs/http2/vcenter.lab.local/443/ds/vapi,
ssoDomain=vsphere.local]: Provider method implementation threw unexpected exception: Invalid token
```

Root Cause:

The vCenter Security Token Service (STS) was returning invalid tokens. Investigation revealed that the STS_INTERNAL_SSL_CERT_VECS (VMware Endpoint Certificate Store) store was missing entirely. Attempting to list the store returned Error 4312 (ERROR_OBJECT_NOT_FOUND). However, the actual STS

signing certificates were present and valid in LDAP (vmdir).

Evidence -- VECS Store Missing:

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store STS_INTERNAL_SSL_CERT --text | grep -E "(Not Before|Not After|Subject)"
vecs-cli failed. Error 4312: Possible errors: LDAP error: Unknown (extension) error Win Error: Operation failed with error ERROR_OBJECT_NOT_FOUND (4312)
```

Evidence -- VECS Store List:

```
/usr/lib/vmware-vmafd/bin/vecs-cli store list
```

Output showed the following stores. STS_INTERNAL_SSL_CERT was NOT present:

```
MACHINE_SSL_CERT, TRUSTED_ROOTS, TRUSTED_ROOT_CRLS, machine, vsphere-webclient, vpxd, vpxd-extension, hvc, data-encipherment, SMS, APPLMGMT_PASSWORD, wcpssc
```

Evidence -- MACHINE_SSL_CERT Valid:

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store MACHINE_SSL_CERT --text | grep -E "(Not Before|Not After|Subject)"
```

```
Not Before: Feb 7 00:26:39 2026 GMT Not After : Feb 7 12:26:39 2028 GMT Subject: CN=vcenter.lab.local, C=US
```

Evidence -- STS Certs Valid in LDAP:

```
/usr/lib/vmidentity/support/sts-cert-support.sh --help
```

Output showed two certificate entries in LDAP:

```
❖ TenantCredential-1 (ssoserverSign): Not Before Feb 7, 2026 -- Not After Feb 2, 2036. Subject: CN=ssoserverSign. Valid for 10 years.
❖ TrustedCertChain-1 (CA): Not Before Feb 4, 2026 -- Not After Feb 2, 2036. Subject: CN=CA, DC=vsphere, DC=local. Valid for 10 years.
```

Conclusion:

The STS signing certificates themselves were not expired (valid until 2036). The MACHINE_SSL_CERT was also valid (until 2028). The root cause was the missing STS_INTERNAL_SSL_CERT VECS store, which prevented the local STS service from properly signing and validating tokens. The STS service required a restart to recover.

fixsts.sh Not Available:

The standard VMware KB fix (fixsts.sh) was not present on this vCenter appliance. The search confirmed:

```
find /usr/lib -name "fixsts*" 2>/dev/null (no results)
```

Available scripts in /usr/lib/vmidentity:

```
/usr/lib/vmidentity/install/sts-start-script.sh /usr/lib/vmidentity/install/configure-build.sh
/usr/lib/vmidentity/install/cleanup-old-jit-fsps.sh
/usr/lib/vmidentity/install/standalone-install.sh /usr/lib/vmidentity/support/sts-cert-support.sh
```

6. Resolution Steps

6.1 Phase 1: Connectivity Verification

Before modifying any credentials or services, network connectivity was verified from the VCF Operations collector appliance to all dependent systems.

Step 1: Verify SDDC Manager reachability

Executed from the collector appliance CLI (root@collector):

```
curl -k https://sddc-manager.lab.local
```

Result:

```
<html><head><title>301 Moved Permanently</title></head><body>...nginx...</body></html>
```

The HTTP 301 redirect from nginx confirmed SDDC Manager's web server was responding. Connectivity verified.

Step 2: Verify NSX Manager reachability

```
curl -k https://nsx-vip.lab.local
```

Result: Empty response (no output). NSX Manager web server responded with an empty page. Connectivity verified.

Step 3: Test NSX API authentication

```
curl -k https://nsx-vip.lab.local/api/v1/cluster/status
```

Result:

```
{"module_name":"common-services","error_message":"The credentials were incorrect or the account specified has been locked.","error_code":403}
```

Confirmed: NSX Manager is reachable but rejecting authentication. This is a credential problem, not a connectivity problem.

6.2 Phase 2: NSX Admin Credential Update

Step 1: Verify NSX admin account is not locked

Connected to NSX Manager CLI and checked admin account status:

```
nsx-node1> get user admin status
```

Result:

```
Tue Mar 31 2026 UTC 12:06:08.446 Full name: Username: admin Status: ACTIVE
```

Admin account was ACTIVE (not locked). The issue was an incorrect or missing password in VCF Operations, not a locked account.

Step 2: Update admin credential in VCF Operations

- ❖ Navigated to Administration > Integrations > Accounts in VCF Operations.
- ❖ Clicked the three-dot menu next to nsx-vip.lab.local.
- ❖ Clicked Edit.
- ❖ Updated the NSX admin username and password to the current credentials.
- ❖ Clicked Test Connection -- connection succeeded.
- ❖ Clicked Save.

After saving, the admin connection was restored. However, the audit and root accounts remained disconnected.

6.3 Phase 3: NSX Audit Account Password Reset

Step 1: Check audit account status

```
nsx-node1> get user audit status  
Status: PASSWORD_EXPIRED
```

Step 2: Reset audit password

```
nsx-node1> set user audit password New password: Confirm new password:
```

New password was entered and confirmed.

Step 3: Extend password expiration

```
nsx-node1> set user audit password-expiration 9999
```

Set expiration to 9999 days to prevent recurrence.

Step 4: Verify status (still expired)

```
nsx-node1> get user audit status  
Tue Mar 31 2026 UTC 12:15:04.927 Full name: Username: audit Status: PASSWORD_EXPIRED
```


Despite the password reset, the account still showed PASSWORD_EXPIRED. The password change from the admin CLI session did not fully activate the account.

Step 5: Check root account status

```
nsx-node1> get user root status
% Operation is not valid for root user
```

The root user does not support the get user status command on NSX Manager. Root is managed differently from other local accounts.

Step 6: Force password activation via interactive SSH

Exited the NSX Manager CLI session and returned to the collector appliance shell. Then initiated an SSH login as the audit user:

```
root@collector [ ~ ]# ssh audit@nsx-vip.lab.local
```

The interactive SSH login forced the password activation workflow to complete. After logging in and setting the password, the audit account became active.

Step 7: Reset root password

The root account password was reset using the same approach (set user root password from NSX CLI, followed by interactive SSH login if needed).

Step 8: Confirm all accounts active

All three NSX accounts (admin, audit, root) confirmed active and able to authenticate.

NOTE: The initial attempt to use 'get user admin audit-login' to check for a locked account failed because that command does not exist on this NSX version. The correct commands are: 'get user <username> password-expiration', 'get user <username> ssh-keys', and 'get user <username> status'.

6.4 Phase 4: VCF Operations Credential Configuration

Step 1: Add NSX credentials to VCF Operations

The NSX credentials were not previously stored in VCF Operations -- they were missing entirely, not just incorrect.

- ◆ Navigated to Administration > Integrations > Accounts.
- ◆ Clicked the CREDENTIALS button (top right of the Integrations page).
- ◆ Added credentials for the NSX admin, audit, and root accounts using the newly reset passwords.
- ◆ Saved each credential entry.

Step 2: Wait for collection cycle

VCF Operations collects data every 5 minutes. After adding the credentials, waited 5-10 minutes for the next collection cycle to verify that the NSX adapter reconnected.

6.5 Phase 5: vCenter STS Service Recovery

Step 1: SSH to vCenter Server

```
ssh root@vcenter.lab.local
```

Step 2: Check trusted certificates

```
/usr/lib/vmware-vmafd/bin/dir-cli trustedcert list --login administrator@vsphere.local --password '<SSO_PASSWORD>'
```

Note: The SSO password contained special characters (! exclamation marks). The password had to be wrapped in single quotes to prevent bash from interpreting the ! characters as history expansion. Using double quotes or no quotes resulted in '-bash: !0909: event not found'.

Result:

```
Number of certificates: 1 CN(id): BF0F4519BDF9129782C64511942B9D02F5D99A19 Subject DN: CN=CA,
```

DC=vsphere, DC=local, C=US, ST=California, O=vcenter.lab.local, OU=VMware Engineering CRL present:

yes

Step 3: Check STS service status

```
vmmon-cli --status sts
```

The STS service was found to be stopped. It was started:

```
vmmon-cli --start sts
```

Step 4: Verify all services running

```
service-control --status --all
```

Running services (confirmed):

```
applmgmt lookupsvc lwsmd observability observability-vapi pschealth vc-ws1a-broker vlcm vmafd  
vmcad vmdird vmware-analytics vmware-certificateauthority vmware-certificatemanagement  
vmware-cis-license vmware-content-library vmware-eam vmware-envoy vmware-envoy-hgw  
vmware-envoy-sidecar vmware-envoy-system-proxy vmware-hvc vmware-infraprofile vmware-perfcharts  
vmware-postgres-archiver vmware-rhttpproxy vmware-sca vmware-sps vmware-std vmware-topologysvc  
vmware-trustmanagement vmware-updatemgr vmware-vapi-endpoint vmware-vdtc vmware-vmmon  
vmware-vpostgres vmware-vpxd vmware-vpxd-svcs vmware-vsan-health vmware-vsm vsphere-ui vstats  
vtsdb wcp
```

Stopped services (expected):

```
vmcam vmware-imagebuilder vmware-netdumper vmware-rbd-watchdog vmware-vcha
```

All critical services were running. The stopped services (vmcam, imagebuilder, netdumper, rbd-watchdog, vcha) are optional/unused in this environment.

Step 5: Check STS_INTERNAL_SSL_CERT VECS store

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store STS_INTERNAL_SSL_CERT --text | grep -E "(Not  
Before|Not After|Subject)"  
vecs-cli failed. Error 4312: Possible errors: LDAP error: Unknown (extension) error Win Error:  
Operation failed with error ERROR_OBJECT_NOT_FOUND (4312)
```

The STS_INTERNAL_SSL_CERT VECS store was missing. This was confirmed by listing all VECS stores:

```
/usr/lib/vmware-vmafd/bin/vecs-cli store list
```

STS_INTERNAL_SSL_CERT was NOT in the output.

Step 6: Verify MACHINE_SSL_CERT is valid

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store MACHINE_SSL_CERT --text | grep -E "(Not  
Before|Not After|Subject)"  
Not Before: Feb 7 00:26:39 2026 GMT Not After : Feb 7 12:26:39 2028 GMT Subject:  
CN=vcenter.lab.local, C=US
```

MACHINE_SSL_CERT was valid (expires Feb 2028). Not the cause of the issue.

Step 7: Verify STS certs in LDAP

```
/usr/lib/vmidentity/support/sts-cert-support.sh --help
```

Output confirmed two certificate entries in LDAP (vmdir), both valid until 2036. See Section 5.3 for full certificate details.

Step 8: Attempt to locate fixsts.sh

```
find /usr/lib -name "fixsts*" 2>/dev/null
```

No results. The fixsts.sh script (referenced in VMware KB articles for STS certificate repair) was not present on this vCenter appliance.

```
find /usr/lib/vmidentity -name "*.sh" 2>/dev/null
```

Available scripts listed (see Section 5.3). The sts-cert-support.sh script was used for certificate verification but does not regenerate certificates.

Step 9: Restart STS service

Initial attempts with incorrect service names failed:

```
root@vcenter [ ~ ]# vmon-cli --restart vmware-stsd Restart service request failed. Error: Invalid
input data root@vcenter [ ~ ]# vmon-cli --restart vmware-sts Restart service request failed.
Error: Invalid input data
```

The correct service name was identified by listing services:

```
vmon-cli --list | grep -i sts
```

Output: sts

Restarted with the correct service name:

```
vmon-cli --restart sts
```

7. Commands Reference

Complete list of every command executed during this troubleshooting session, in order of execution.

#	Where	Command	Purpose
1	Collector	curl -k https://sddc-manager.lab.local	Verify SDDC Manager reachability
2	Collector	curl -k https://nsx-vip.lab.local	Verify NSX Manager reachability
3	Collector	curl -k https://nsx-vip.lab.local/api/v1/cluster/status	Test NSX API authentication
4	NSX CLI	get user admin status	Check if admin account is locked
5	NSX CLI	set user audit password	Reset expired audit password
6	NSX CLI	set user audit password-expiration 9999	Extend audit password expiration
7	NSX CLI	get user audit status	Verify audit account status
8	NSX CLI	get user root status	Check root status (failed: not root)
9	Collector	ssh audit@nsx-vip.lab.local	Force password activation via login
1	NSX CLI	set user root password	Reset root password
0		/usr/lib/vmware-vmafd/bin/dir-cli trustedcert list --login	List trusted certificates
1	vCenter	administrator@vsphere.local --password '<PASSWORD>'	Check STS VECS store (Error 4312)
1		vmon-cli --status sts	List all VECS stores
1	vCenter	vmon-cli --start sts	Verify machine SSL cert validity
2		service-control --status --all	Verify STS certs in LDAP
1	vCenter	/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store	Search for fixsts.sh script
3		STS_INTERNAL_SSL_CERT --text	List available identity scripts
1	vCenter	/usr/lib/vmware-vmafd/bin/vecs-cli store list	Restart STS (failed: wrong service name)
4		/usr/lib/vmware-vmafd/bin/vecs-cli entry list --store	Restart STS (failed: wrong service name)
1	vCenter	MACHINE_SSL_CERT --text	Restart STS (failed: wrong service name)
5		/usr/lib/vmidentity/support/sts-cert-support.sh --help	
1	vCenter	find /usr/lib -name "fixsts*" 2>/dev/null	

```

6          find /usr/lib/vmidentity -name "*.sh" 2>/dev/null
1 vCenter
7          vmon-cli --restart vmware-stsd
1 vCenter vmon-cli --restart vmware-sts
8
1 vCenter
9
2 vCenter
0
2 vCenter
1
2 vCenter
2
2 vCenter vmon-cli --list | grep -i sts Find correct STS service name
3          vmon-cli --restart sts Restart STS service (correct name)
2 vCenter
4

```

8. Verification and Current Status

Issue	Status	Verification Method
NSX admin credential page shows admin	RESOLVED -- Admin password updated in VCF Operations. NSX adapter reconnected for admin.	VCF Operations Integrations connection active.
NSX audit account expired returns ACTIVE on NSX CLI.	RESOLVED -- Password reset, expiration extended to 9999 days, account activated via interactive SSH login.	get user audit status
NSX root account credentials in VCF Ops	RESOLVED -- Password reset and account activated.	Root able to authenticate.
VCF adapter cascade Integrations > CREDENTIALS warnings	RESOLVED -- All three credentials (admin, audit, root) added to VCF Operations credential store.	Administration > shows all NSX entries.
vCenter STS Invalid Token after next collection	PENDING VERIFICATION -- Expected to clear within 5-10 minutes after NSX adapter reconnects.	Refresh Integrations page cycle. All adapters should
show Collecting. STS_INTERNAL_SSL_CERT store https://vcenter.lab.local/	IN PROGRESS -- STS service restarted. Awaiting confirmation that vSphere Client authentication functions normally.	Log into vSphere Client at ui. Verify no 'Invalid
token' errors. store list -- check if appears. If not,	OPEN -- VECS store still missing. STS restart may or may not recreate it. Monitor for recurrence.	After restart: vecs-cli STS_INTERNAL_SSL_CERT may require certificate-reset.

9. Preventive Recommendations

# Recommendation Priority	Details
1 Extend NSX password set Critical	Run on all NSX Manager nodes: set user admin password-expiration 9999
expiration for all service accounts	user audit password-expiration 9999 This prevents future authentication failures due to expired passwords. The root account does not support

this command and must be managed separately.

2 Monitor credential status Create a scheduled check (weekly) to verify all adapter instances on the High

in VCF Operations Integrations page show Collecting status. Set up an email alert or dashboard widget for adapter health.

3 Document all service High Maintain a secure password vault entry for all VCF component service account passwords accounts: NSX admin/audit/root, vCenter SSO administrator, SDDC Manager admin, ESXi root. Update the vault whenever passwords are rotated.

4 Monitor vCenter store High Schedule a monthly check of all VECS certificate stores. Command: for certificate expiration in `$(vecs-cli store list); do echo "=== $store ==="; vecs-cli entry list --store "$store" --text 2>/dev/null | grep 'Not After'; done` Alert if any certificate expires within 90 days.

5 Investigate missing certificate Medium The missing VECS store may indicate a previous failed upgrade or STS_INTERNAL_SSL_CE rotation. If the STS restart does not recreate the store, run:

RT store /usr/lib/vmware-vmca/bin/certificate-manager Option 8 (Reset all certificates).

This is a disruptive operation -- schedule during a maintenance window.

6 Standardize vmon-cli Low Document the correct service names for vmon-cli. Common confusion: the service names STS service name is 'sts', not 'vmware-std' or 'vmware-sts'. Use 'vmon-cli --list' to verify names before restart.

7 Use single quotes for the Low When passing passwords containing ! or other bash special characters on the command line, always use single quotes: `--password 'MyP@ss!word'`. Double quotes and unquoted passwords will cause bash history expansion errors.

10. Affected Component Summary

Component Status	Issue	Root Cause	Resolution
NSX Manager Resolved (nsx-vip.lab.local)	VCF Operations adapter Warning -- 403 on API	Admin credentials missing from VCF Operations credential store	Added correct credentials to VCF Operations
NSX audit Resolved account + days	PASSWORD_EXPIRED status	Password expiration policy reached; CLI reset insufficient without interactive login	Password reset + interactive SSH login to force activation expiration extended to 9999
NSX root Resolved account	Disconnected from VCF Operations	Password expired or incorrect; 'get user root status' not supported	Password reset + credentials added to VCF Operations
VCF Operation	Warning cascade on		Resolved NSX adapter; cascade

Pending s Verification adapters (all 5)	all VMware Cloud Foundation adapters	NSX child adapter failure cascaded to parent and sibling adapters	clears automatically
vCenter Server In Progress (vcenter.lab.local)	Invalid token -- STS authentication failure in vSphere Client	STS_INTERNAL_SSL_CERT VECS store missing; STS service needed restart	STS service restarted via vmon-cli --restart sts
vCenter VECS Open STS_INTERNAL_SSL_CERT	VECS store missing (Error 4312)	Unknown -- possibly failed upgrade or previous certificate rotation issue	STS restart; if store not recreated, certificate- Option 8 required
-- End of Root Cause Analysis Report --			

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